



On the impact of mechanistic model quality and data availability in hybrid model development

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ARTICLE INFO

Keywords:

Hybrid models
Mechanistic models
Data-driven models
Data availability
Random forest
Ion-exchange chromatography

ABSTRACT

This work presents a methodology for evaluating the effectiveness of hybrid modelling under varying conditions of mechanistic model quality and information available for model training, that is typically expressed as the amount of data available. While hybrid models – which integrate mechanistic and data-driven components – have gained significant attention in process systems engineering, their advantages over purely mechanistic or data-driven alternatives remain inadequately quantified. We address this gap by investigating two critical factors: (i) the impact of mechanistic model fidelity on hybrid model performance, and (ii) the influence of calibration dataset size on prediction accuracy. Our methodology is validated through an *in-silico* case study of baker's yeast cultivation and a real-world industrial application of ion-exchange chromatography in biopharmaceutical manufacturing. Results demonstrate that hybrid models consistently outperform purely mechanistic and data-driven approaches when the mechanistic component captures fundamental process behaviours, even with structural simplifications. Notably, hybrid models maintain superior predictive capability in extrapolative scenarios; however, when mechanistic knowledge is severely limited and insufficient information is available for compensation, hybridisation benefits diminish substantially. The work provides quantitative guidance for practitioners to determine when hybrid modelling represents a justified investment of modelling resources in process engineering applications.

1. Introduction

In recent years, a variety of modelling approaches have gained significant attention to support the design and optimisation of engineering processes across diverse industrial sectors (Zhang et al., 2019). The growing interest in adopting mathematical models is motivated by the increasing demand for more efficient and reliable approaches to process development and decision-making, especially in contexts where experimental campaigns are costly, time-consuming, or limited by operational constraints (Wang et al., 2024). The level of knowledge and understanding of the process, as well as the amount of information required for calibration and validation of a model at a given stage of process development, can vary significantly, affecting the selection of the modelling method (Chen and Ierapetritou, 2020; Zendejboudi et al., 2018).

Depending on the extent of the underlying physics that is captured by

the model equations, mathematical models are generally categorized into three main types (Bonvin et al., 2016; Moser et al., 2020): mechanistic models (also referred to as knowledge-driven, first-principles, or white-box models), data-driven models (often called empirical or black-box models), and hybrid models (also known as semi-empirical or grey box models). One of the primary benefits of mechanistic models is their ability to enhance understanding of products and processes by explicitly describing the physical, chemical, and biological properties of the system under investigation. They typically require few calibration experiments and provide superior extrapolation capabilities compared to data-driven approaches. However, deriving their equations may be complex, as it involves identifying the exact underlying physical phenomena that characterize the process under analysis, and demands more time, resources, and modelling effort, often requiring high computational burden (Solle et al., 2017). In cases of limited knowledge and

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process understanding, a mechanistic modelling approach may be insufficient to reliably enable simulation-based development and optimisation, often resulting in process-model mismatch, i.e., a discrepancy between predictions and experimental data.

On the other hand, empirical models only rely on available historical data. The main advantages include their ability to capture correlation structures within data with no need of detailed understanding of the underlying physics, rapid development timelines, and reduced computational demands, particularly when they offer simplified representations of highly complex mechanistic models (Reis and Saraiva, 2019; Rendall and Reis, 2018). Nonetheless, they typically require extensive calibration datasets, which may be limited or inconsistent in terms of structure and quality. Furthermore, generalization and interpretability can be more challenging since their performance may strongly reduce when applied to explorative scenarios (Sansana et al., 2021).

When tackling real-world problems, all available sources of information (i.e., physical understanding of the system and historical process data) should be leveraged. In this context, hybrid modelling strategies are effective alternatives aiming to integrate the physical knowledge embedded in the available mechanistic model with empirical data (von Stosch et al., 2014). As a result, hybrid configurations preserve core insight into the underlying physical process mechanisms while extracting information from available data-driven patterns that cannot be captured by first-principles knowledge, thereby enhancing both predictive accuracy and interpretability (Sansana et al., 2021). Additionally, hybrid models can contribute to reducing the overall time required for model development. In general, extending a mechanistic model to include more detailed knowledge can be a time-consuming task (Schweidtmann et al., 2024). By leveraging the flexibility of data-driven components, part of the knowledge not explicitly captured in the mechanistic equations can instead be extracted by appropriately modelling the available process data, thereby accelerating the construction of reliable process models.

Over the last years, various studies have explored the potential of hybrid models across different domains. For instance, Schäfer et al. (2019) used artificial neural networks combined with differential equations to model rectification columns, and the model was later used for real-time applications (Lüthje et al., 2021). Bui et al. (2021) proposed a hybrid model configuration combining a fundamental adiabatic reactor model with an empirical partial least-squares regression to monitor and predict the catalyst lifetime of an operating industrial reactor. Shah et al. (2022) demonstrated the high predictive capability of a deep neural network-based hybrid model for an industry-scale fermentation process, outperforming a purely mechanistic kinetic model. Polak et al. (2023) developed a hybrid modelling approach to evaluate the effects of raw material variations and process parameter fluctuations on product quantity and quality in a flow chemistry process for small molecules manufacturing. Xie et al. (2024) employed a neural network to compensate for the unmodeled dynamics of the first-principles model on an industrial nitration process under load-changing scenarios. Mahanty et al. (2025) proposed a hybrid model to predict diesel fuel removal dynamics, with the instantaneous removal rate modelled using an artificial neural network. Michalopoulou and Papathanasiou (2025) integrated artificial neural networks with process knowledge to describe the nonlinear dynamics of a chromatographic separation process. Caspari et al. (2025) provided evidence that hybrid models remain effective in data-scarce scenarios, highlighting their robustness with limited information. Steinacker et al. (2025) explored the potential of hybrid models based on different neural networks and compared the performance to typically used mechanistic models to predict atypical trajectories of hematotoxicity damage to chemotherapy patients. Wang et al. (2025) demonstrated the improved performances of physics-data-driven hybrid models to predict accurately the friction torque in hydraulic motors.

However, not only do hybrid models require a careful integration of system knowledge and historical data, but they also necessitate a deep

understanding of when and how each component of the model (i.e., the mechanistic and data-driven ones) can contribute to the overall hybrid performance (Rogers et al., 2023; Bradley et al., 2022). Developing a comprehensive framework to systematically construct and integrate these components remains challenging (Shah et al., 2024). This represents a clear need for further development to ensure that hybrid approaches can be more systematically and effectively applied in process engineering (Schweidtmann et al., 2024). Most published contributions analyzing how the quality each component or the limited amount of information available for model training impact on the overall hybrid model performance are based on synthetic or simulated datasets, with a notable lack of studies addressing real-world processes in a generalizable framework. These works are also generally limited to specific architectures (Narayanan et al., 2022).

Our methodology is designed for cases in which models are developed for offline analysis and optimization purposes, such as for process design, scale-up and simulation. While several studies have explored hybrid modelling architectures and performance metrics, most contributions do not quantify the relative impact of mechanistic accuracy and data availability on the predictive reliability of hybrid models. This work specifically addresses this gap by analysing how these two factors interact and affect model robustness under both *in-silico* and industrial conditions.

Our investigation encompasses both *in-silico* and real-world case studies. Specifically, an *in-silico* study conducted on a fed-batch reactor system for baker's yeast production is used to address key challenges, such as the limited predicative capability of available mechanistic models and the paucity of data, and questions the effectiveness of hybrid modelling under such constraints. An additional *in-silico* case study concerning urethane synthesis is presented in the Supplementary Material. Finally, a real-world case study concerning ion-exchange chromatography in biopharmaceutical manufacturing aims at providing insights concerning typical challenges in industrial applications.

The paper is organized as follows. In Section 2, we describe the general methodology for the analysis and construction of suitable hybrid modelling configurations. In Section 3, we give details about the *in-silico* and real-world case studies, respectively. Results are presented and critically discussed in Section 4. Final remarks conclude the manuscript.

2. Methodology

2.1. General methodology

The methodology is structured into a sequence of steps, each of them contributing to the systematic construction and evaluation of suitable hybrid models (Fig. 1). Here, we aim to assess: (i) the impact of different levels of complexity in the mechanistic component, and (ii) the effect of the amount of information available on the overall hybrid model performance. These aspects are crucial to evaluate the robustness and generalization capabilities of the methodology under various practical scenarios. The methodological steps are outlined in Fig. 1.

2.1.1. Step #1 – mechanistic model calibration and validation

We start from an available mechanistic model, which is assumed to provide an approximate representation of the system under investigation, based on the modeler's prior knowledge. In mathematical terms, the relationship between the process inputs and outputs can be described by the following set of differential and algebraic equations (DAEs):

$$\begin{cases} \mathbf{f}(\mathbf{x}(t), \dot{\mathbf{x}}(t), \mathbf{u}(t), \boldsymbol{\theta}, t) = \mathbf{0} \\ \hat{\mathbf{y}}(t) = \mathbf{h}(\mathbf{x}(t)) \end{cases}, \quad (1)$$

where $\mathbf{x}(t)$, $\mathbf{u}(t)$, and $\boldsymbol{\theta}$ are (respectively) the vector of time-dependent state variables, the vector of controls or input variables that can be manipulated during an experiment, and the vector of the model parameters; $\hat{\mathbf{y}}$ is the vector of predicted output variables, while $\mathbf{f}$ is the

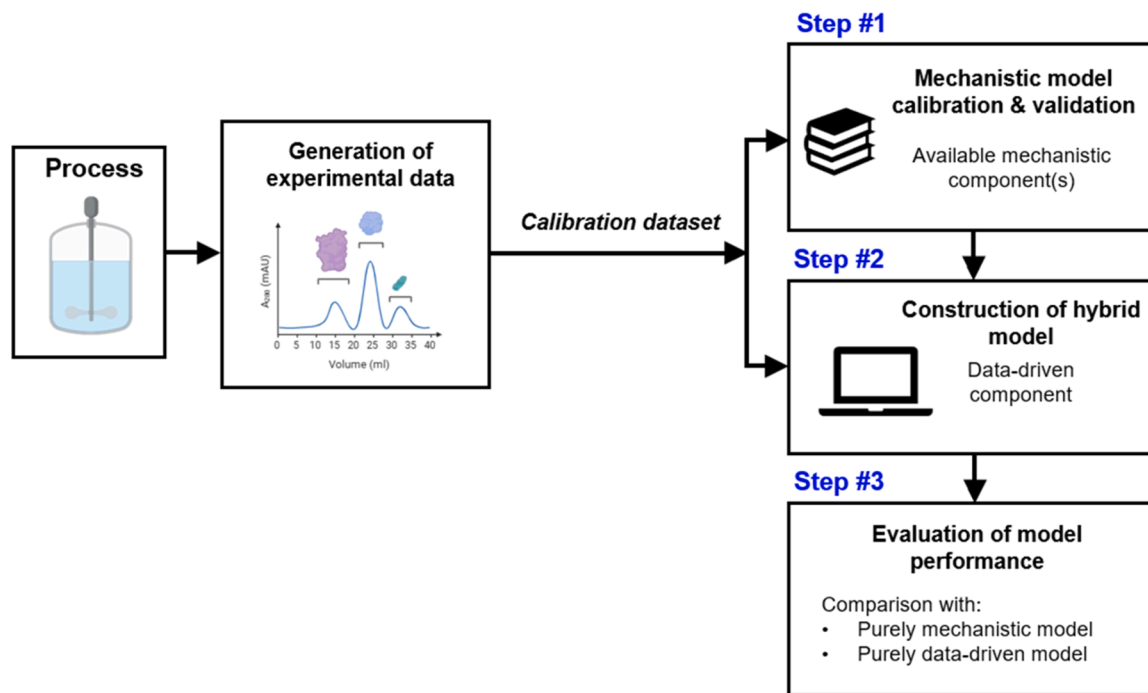


Fig. 1. Steps to build and evaluate the benefits of a hybrid model.

vector field that defines the system of DAEs governing the process dynamics; $\mathbf{h}$ is the output map, i.e., the vector-valued function that maps the state variables to the observable output quantities.

Assuming that the model is practically identifiable (Miao et al., 2011) – i.e., all its parameters can be estimated from measured data – the first step consists in calibrating the mechanistic model using a subset of the available experimental data, either generated *in-silico* or from a real experiment. In this work, parameter estimation is performed via nonlinear least-squares regression, by minimizing the sum of squared errors (SSE), ϕ^{LS} , between the measured outputs $\mathbf{y}_{exp,i}$ and the model predictions $\hat{\mathbf{y}}_i$ across all available experiments and time points (Sorenson, 1980):

$$\phi^{LS} = \sum_{i=1}^{N_{sp}} \left[\left(\mathbf{y}_{exp,i} - \hat{\mathbf{y}}_i \right)^T \left(\mathbf{y}_{exp,i} - \hat{\mathbf{y}}_i \right) \right], \quad (2)$$

where N_{sp} is the total number of samples. Note that other estimators can be adopted, such as the weighed least-squares, or the maximum likelihood estimator (Bard, 1974).

The remaining experiments are reserved for validation of the model performance, and, if possible, should cover both conditions within the calibration design space and extrapolative cases, provided they are still representative of the operational domain of interest. This validation strategy allows to assess both model accuracy under familiar conditions, but also its generalization capabilities when applied to unseen regions of the experimental domain.

2.1.2. Step #2 – construction of hybrid model

While mechanistic models offer a structured and interpretable description of the underlying physical and chemical phenomena, they often suffer from structural inaccuracies due to simplifications, unmodeled dynamics, or missing mechanisms. To account for process-model mismatch, a hybrid modelling structure that integrates a data-driven component into the mechanistic framework can be used. Specifically, we propose an architecture where the mechanistic component provides initial predictions of the states and outputs of the system (Sansana et al., 2021); The data-driven component takes the mechanistic component outputs, along with the experimental inputs available, and is

constructed to model the estimation error of the mechanistic component, thus compensating for the observed mismatch (Fig. 2).

The proposed architecture differs from the typical serial or parallel configurations reported in the literature (Schäfer et al., 2020; Shah et al., 2024). It combines elements of both approaches: similar to parallel structures, the data-driven component corrects the output of the mechanistic model by compensating for its systematic deviations. At the same time, it shares a feature with serial structures, since the data-driven model leverages the information provided by the mechanistic model itself. This results in a formulation that benefits from both strategies, without being a conventional instance of either.

Random forest (RF) models are employed as data-driven components (Breiman, 2001). They make output predictions by combining the outcomes from a sequence of regression decision trees, called forest; each tree depends on the values of a random vector sampled independently and with the same distribution for all trees in the forest. The output value for the RF is simply the average value of the outputs for every decision tree in the forest.

The selection of RFs as data-driven component is motivated by their ability to capture nonlinear relationships, and because they tend to perform well in low-data regimes while exhibiting relatively low risk of overfitting (He and Garcia, 2009). However, it should be noted that the signal predicted by RFs may appear noisy, since the implementation does not account for temporal autocorrelation. As a result, the predictions may lack smoothness, and a filtering step was applied to dampen these fluctuations (Blackledge, 2006). The moving average $\tilde{y}_i$ of a series y_i over a window size ω is defined as:

$$\tilde{y}_i = \frac{1}{\omega} \sum_{j=-k}^k y_{i+j}, \quad (3)$$

where $\omega = 2k + 1$ is the window size, and k is the number of data points on either side of y_i . The window size ω should be large enough to smooth high-frequency noise, while at the same time retain the significant features of the signal, i.e., the predicted output.

The key hyperparameter of a RF is the number of trees in the forest; in this work, the optimal value is determined by minimizing the root-mean squared error through cross-validation (He and Garcia, 2009).

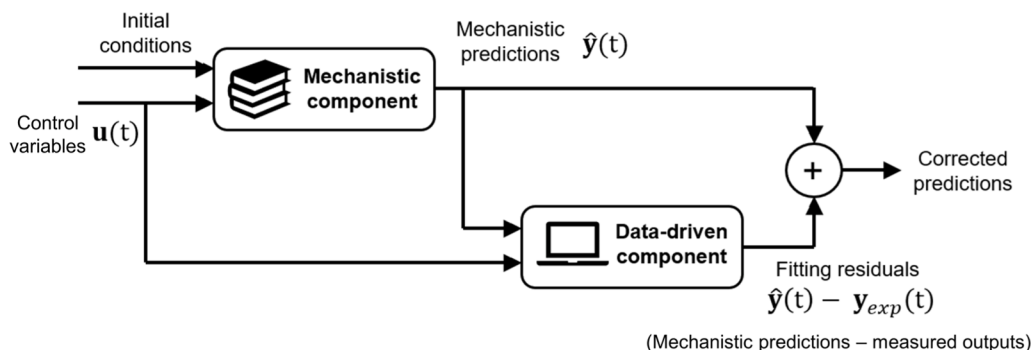


Fig. 2. Proposed hybrid model configuration.

Alternative optimisation strategies are also possible, such as a grid-search approach (Bergstra et al., 2013).

RFs demonstrated very good performance in the case studies considered in this work. Nonetheless, other machine learning algorithms – such as artificial neural networks (Haykin, 2009), Gaussian processes (Rasmussen and Williams, 2005), partial least square (PLS) regression models (Geladi and Kowalski, 1996) or support vector machines (Drucker et al., 2002) – may also be considered, depending on the characteristics of the system and the availability of data.

In the construction of the hybrid model configuration, the data-driven component is the only one subject to parameter estimation (i.e., the decision rules at each tree node). The mechanistic model parameters, which have been calibrated beforehand (see Section 2.1.1), are kept fixed. This defines our sequential approach, where the mechanistic component is first calibrated independently, and subsequently used as the static backbone of the hybrid configuration. While this sequential approach may not guarantee global optimality in terms of overall model performance (Schweidtmann et al., 2024), it is deliberately adopted here for two reasons: (i) as the mechanistic component is assumed to represent the best available knowledge, it serves as fixed backbone of the hybrid model structure, and (ii) avoiding re-calibration of the mechanistic part allows for a clean and interpretable comparison across different scenarios, isolating the contribution and compensatory effect of the data-driven component without the influence of re-estimating the mechanistic parameters.

A simultaneous approach may lead to global optimality in terms of prediction accuracy. However, such an approach was not considered since it also introduces additional complexity: (i) a risk of losing parameter interpretability, as the optimization may compensate mechanistic inaccuracies through parameter values with no physical meaning; (ii) higher computational cost due to the enlarged parameter space.

2.1.3. Step #3 – evaluation of model performance

The final step focuses on evaluating whether the integration of the data-driven component into the mechanistic component results in a measurable improvement over purely mechanistic or purely data-driven models, and, if so, to what extent this improvement is achieved. To this end, the hybrid model performance should be compared against that of the standalone mechanistic and empirical models. The quality of fitting is measured using the mean absolute error (MAE):

$$MAE = \frac{1}{N_{sp}} \sum_{i=1}^{N_{sp}} |y_i - \hat{y}_i| \quad (4)$$

2.2. Computational details

All activities were performed on an Intel Core I7-11850H CPU@2.50 GHz processor with 64.0 GB RAM or an Intel Core I9-13900H@2.60 GHz processor with 64 GB RAM. All models were implemented as custom Python scripts (version 3.12).

The generation of artificial datasets for the *in-silico* case studies required only a few minutes per case. In contrast, parameter estimation for mechanistic models proved to be more computationally intensive, ranging from tens of minutes (e.g., baker's yeast growth and chromatography studies) to several hours (for the case study concerning urethane production process presented in the Supplementary Material), depending on model complexity and the amount of experimental data used for calibration. Parameter estimation of hybrid and purely data-driven models typically required between a few minutes and several tens of minutes, depending on the size of the calibration dataset.

3. Case studies

Two case studies will be presented in the following sections. The first case study involves a simulated fed-batch bioreactor and serves as an *in-silico* testbed for exploring how hybrid model performance is affected by mechanistic component simplification and amount of information available.

The second case study focuses on a real-world ion-exchange chromatography process, aiming to test the applicability and robustness of the proposed methodology in the presence of complex physicochemical mechanisms and industrial data constraints.

An additional *in-silico* case study of increasing complexity is discussed in the Supplementary Material.

3.1. Baker's yeast growth

3.1.1. Process

The first process is yeast cultivation in a fed-batch bioreactor. We aim at building a model that predicts the dynamics of yeast and substrate concentrations. This is an *in-silico* case study; the process is assumed to be perfectly described by the following set of mass balance equations (Aiba et al., 1968):

$$\frac{dy_1}{dt} = (r - u_1 - \theta_4)y_1, \quad (5)$$

$$\frac{dy_2}{dt} = -\frac{ry_1}{\theta_3} + u_1(u_2 - y_2), \quad (6)$$

$$r = \frac{\theta_1 y_2}{\theta_2 + y_2} e^{-\frac{y_2}{\theta_5}}, \quad (7)$$

where y_1 and y_2 are (respectively) the biomass and substrate concentrations inside the reactor, u_1 is the dilution factor, u_2 is the substrate concentration in the feed. The number of parameters is 5, and they are assumed to be known and fixed at their nominal values (Table 1).

Datasets are generated by solving the model Eqs. (5)–(7) with parameter values as in Table 1. A variety of experimental conditions is considered by varying manipulated variables and initial conditions across different batch runs, with values randomly drawn from

Table 1

Case study #1 – Baker's yeast growth. Nominal values of model parameters.

Parameter	Units	Value	Physical meaning
θ_1	h^{-1}	0.31	Maximum rate
θ_2	g L^{-1}	0.18	Substrate concentration for half-maximal growth rate
θ_3	–	0.65	Yield
θ_4	h^{-1}	0.25	Reduction in biomass due to other factors
θ_5	g L^{-1}	5.00	Exponential substrate inhibition constant

Table 2

Case study #1 – Baker's yeast growth. Upper and lower bounds for manipulated inputs and initial conditions.

Parameter	Units	Lower bound	Upper bound
u_1	h^{-1}	0.05	0.20
u_2	g L^{-1}	5.00	35.00
y_1^0	g L^{-1}	1.00	10.00
y_2^0	g L^{-1}	0.00	3.00

predefined intervals (Table 2). Specifically, the control variables u_1 and u_2 are treated as piecewise-constant functions with a minimum step duration of 0.5 h, and can vary over 5 steps, while initial concentrations y_1^0 and y_2^0 are random integers. Each simulated batch has a duration of 10 h, with measurements recorded every 10 s. To emulate experimental uncertainty and measurement noise, Gaussian noise with variance $\sigma^2 = 0.02 \text{ g L}^{-1}$ is added to the output profiles of the measured variables y_1 and y_2 (Galvanin et al., 2011). For the theoretical analysis, either all or a subset of the generated datasets are used, allowing the assessment of the impact of amount of available data on hybrid model performance.

The *mechanistic model* is represented by a simplification of Eqs. (5)–(7), intended to reflect the modeler's ability to describe the *process*. Three levels of simplification are considered and discussed in the following, with the aim of investigating the effect of mechanistic component complexity on the overall hybrid model performance. In addition, the effect of the amount of information included in the experimental data will be analysed, i.e., how the amount of data used for the calibration influence model predictive capability.

3.1.2. Mechanistic model #1

In this first simplification, the substrate inhibition effect in the growth rate expression (Eq. (7)) is neglected. The exponential term is removed, resulting in standard Monod-type kinetics:

$$\frac{dy_1}{dt} = (r - u_1 - \theta_4)y_1, \quad (8)$$

$$\frac{dy_2}{dt} = -\frac{ry_1}{\theta_3} + u_1(u_2 - y_2), \quad (9)$$

$$r = \frac{\theta_1 y_2}{\theta_2 + y_2}. \quad (10)$$

This simplification reduces model complexity while still capturing the saturation behaviour of biomass growth with respect to substrate availability. It assumes that inhibition at high substrate concentrations is negligible or irrelevant within the operating conditions.

3.1.3. Mechanistic model #2

In the second simplification, the growth rate expression is simplified to a Monod-type form, and the loss term is removed from the biomass dynamics:

$$\frac{dy_1}{dt} = (r - u_1)y_1, \quad (11)$$

$$\frac{dy_2}{dt} = -\frac{ry_1}{\theta_3} + u_1(u_2 - y_2), \quad (12)$$

$$r = \frac{\theta_1 y_2}{\theta_2 + y_2}. \quad (13)$$

This formulation further reduces model complexity by eliminating the biomass decay term. While the growth rate still depends on substrate concentration, the model assumes ideal conditions with no inhibition or additional biomass loss mechanisms.

3.1.4. Mechanistic model #3

In this model, the yield parameter, θ_3 , is eliminated from the substrate balance, under the assumption of a unit yield. In this scenario the substrate consumption is assumed to be directly proportional to biomass growth without a conversion factor.

$$\frac{dy_1}{dt} = (r - u_1)y_1, \quad (14)$$

$$\frac{dy_2}{dt} = -ry_1 + u_1(u_2 - y_2), \quad (15)$$

$$r = \frac{\theta_1 y_2}{\theta_2 + y_2}. \quad (16)$$

This final simplification represents the most simplified model structure yet retaining a physically meaningful description of the system.

3.2. Ion-exchange chromatography

3.2.1. Process

This case study comes from an industrial biopharmaceutical process. The experimental data concerns monoclonal antibody (mAb) load material with a concentration of 9.12 g L^{-1} , and an average content of heavy and light ends equal to 2.3 % and 1.1 %, respectively. The experimental setup consists in the ion-exchange column Capto SP ImpRes–HiScreen (Cytiva, Marlborough, MA, USA), with bed height of 10 cm, and inner diameter of 0.77 cm. Column bead and interstitial column porosities are equal to 80.6 %, and 38.3 %, respectively; column axial dispersion coefficient, D_{ax} , is $0.22 \text{ mm}^2 \text{ s}^{-1}$. To perform the elution, either a linear gradient or constant salt concentration is used, with the first buffer containing 500 mM sodium acetate, and the second one not containing salts.

Outlet concentration values of mAb for five experimental runs are available (A, B, C, D, E), together with conductivity data and changes in modulator concentration along the elution time span; load velocity, u , is set at 0.22 mm s^{-1} for all runs.

Information on the available datasets used in this study are reported in the [Supplementary Material \(Section S.2\)](#).

3.2.2. Mechanistic model

The physical mechanism underlying ion-exchange chromatography is described through the lumped kinetic model (LKM) (Benedini et al., 2020), which consists of one differential mass conservation equation describing the bulk fluid phase across the column axial coordinate, $z \in (0, L)$, with L representing the column length. The model accounts for convective transport, axial dispersion, and adsorption kinetics, and assumes negligible concentration gradients within the stationary phase. The governing equation is given by:

$$\frac{\partial c_i}{\partial t} = -u \frac{\partial c_i}{\partial z} + D_{ax} \frac{\partial^2 c_i}{\partial z^2} - \frac{1 - \varepsilon_c}{\varepsilon_c} \frac{\partial q_i}{\partial t} \quad \forall z \in (0, L), \quad (17)$$

where c_i is the concentration of solute i in the mobile phase, while q_i represents the solid-phase concentration of species i in the beads. u is the interstitial velocity of material within the unit, D_{ax} is the axial dispersion coefficient, ε_c is the column porosity.

Model equations must satisfy the boundary conditions at the column inlet and outlet. The Danckwerts boundary conditions allow concentration values to fluctuate at the column ends, according to which the

mass flux to the inlet can be expressed as (Nauman and Mallikarjun 1983):

$$uc_{inj,i} = uc_i|_{z=0} + D_{ax} \frac{\partial c_i}{\partial z} \bigg|_{z=0}, \quad (18)$$

where $c_{inj,i}$ is the concentration of species i injected at inlet. At column outlet:

$$\frac{\partial c_i}{\partial z} \bigg|_{z=L} = 0. \quad (19)$$

To describe the relationship between the mobile and stationary phase concentrations, we complement the LKM with a selected adsorption isotherm, as will be discussed in the following.

3.2.2.1. Langmuir adsorption isotherm. The simpler modelling solution used to describe the term $\frac{\partial q_i}{\partial t}$ in Eq. (17) is through the simple Langmuir adsorption isotherm (Guiochon et al., 2006):

$$\frac{\partial q_i}{\partial t} = k_{a,i} c_i q_{max,i} \left(1 - \sum_{j=1}^{N_c} \frac{q_j}{q_{max,j}} \right) - k_{d,i} q_i, \quad (20)$$

where $k_{a,i}$, $k_{d,i}$ are the adsorption and desorption rates of component i , respectively. Each species i among the N_c total components is associated with parameter $q_{max,i}$ that is part of a saturation term allowing the modelling of the total adsorption capacity of the resin. For each species i , three model parameters need to be fitted to experimental data: $k_{a,i}$, $k_{d,i}$, and $q_{max,i}$. It is worth noticing that the choice of the Langmuir adsorption model is a strong simplification, as it does not account for any effects on the adsorption mechanism due to changes in buffer concentration.

Table 3

Case study #1 – Baker's yeast growth. Estimated values of model parameters for the three mechanistic models of different levels of complexity.

Parameter	Units	Mechanistic model #1	Mechanistic model #2	Mechanistic model #3
θ_1	h^{-1}	0.23	0.01	0.04
θ_2	$g L^{-1}$	0.11	0.17	0.32
θ_3	–	0.92	0.06	–
θ_4	h^{-1}	0.30	–	–

3.2.2.2. Asymmetric steric mass action adsorption isotherm. The second modelling strategy used to describe the term $\frac{\partial q_i}{\partial t}$ in Eq. (17) is through the asymmetric steric mass action (ASMA) isotherm (Huuk et al., 2017). The model extends the classical steric mass action theory (Osberghaus et al., 2012), built to take into account the presence of the buffer in the adsorption process, by introducing asymmetry in the association and dissociation mechanisms, as well as non-idealities due to steric hindrance and cooperative interactions:

$$\frac{\partial q_i}{\partial t} = k_{a,i} c_i \bar{q}_1^{\nu_i} \gamma_i - k_{d,i} c_1^{\nu_i} q_i, \quad (21)$$

$$\Lambda = q_1 + \sum \nu_i q_i, \quad (22)$$

$$\bar{q}_1 = q_1 - \sum \sigma_i q_i, \quad (23)$$

$$\gamma_i = \exp(K_{p,i} c_i + K_{1,i} c_1), \quad (24)$$

where $k_{a,i}$ and $k_{d,i}$ are adsorption and desorption rates of component i , ν_i is the characteristic charge of species i , σ_i represents the steric shielding coefficient for species i ; c_1 is the concentration of modulator in the mobile phase, q_1 is the concentration of free binding sites, $\bar{q}_1$ is the effective number of accessible binding sites after accounting for steric hindrance. γ_i is the non-ideality correction factor accounting for cooperative effects and is computed through $K_{p,i}$ and $K_{1,i}$ which are empirical coefficients describing the influence of solute concentrations on the affinity of the binding process. For each species i , six model parameters need to be fitted to experimental data: $k_{a,i}$, $k_{d,i}$, ν_i , σ_i , $K_{p,i}$, and $K_{1,i}$.

4. Results

4.1. Baker's yeast growth

4.1.1. Mechanistic model calibration and validation

To evaluate the effect of the available mechanistic component within the proposed hybrid modelling framework, we begin by calibrating the three mechanistic models that incorporate varying degrees of process knowledge, as described in Section 3.1.

Calibration is performed under a large-data availability scenario, namely using 100 batches for parameter estimation and 10 batches for validation. In validation, we distinguish between in-domain prediction (V-I) – i.e., considering validation batches that follow input conditions

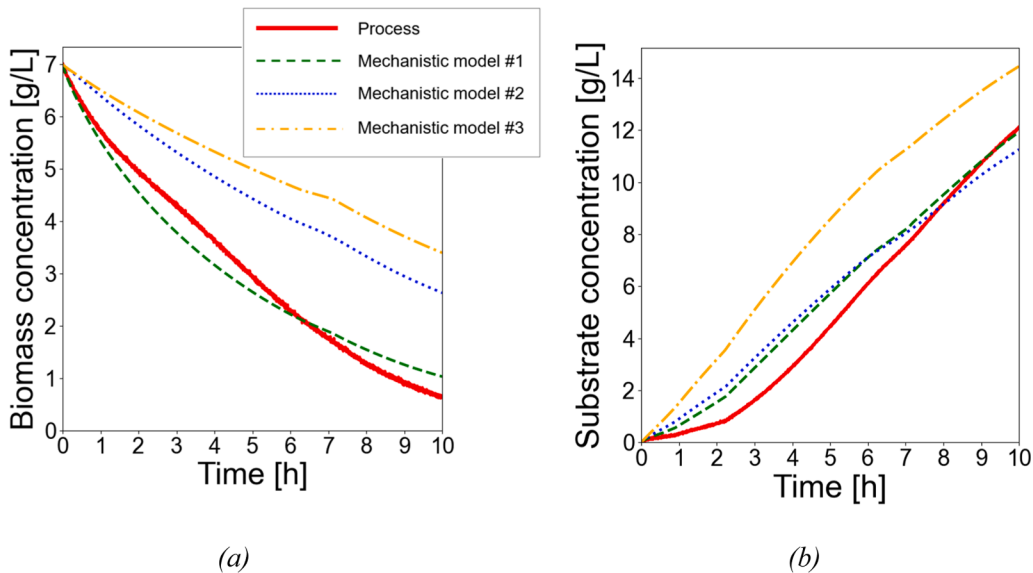


Fig. 3. Case study #1 – Baker's yeast growth. Performance of the three mechanistic models with respect to outputs (a) y_1 , and (b) y_2 on validation batch #1.

Table 4

Case study #1 – Baker's yeast growth. Average MAE for output variables y_1 and y_2 under each scenario for the different mechanistic models considered in this study and the purely data-driven model.

	Mechanistic model #1	Mechanistic model #2	Mechanistic model #3	Data-driven model
Output variable y_1				
MAE in calibration for y_1 [g L ⁻¹]	0.24	1.02	1.34	0.13
MAE V-I for y_1 [g L ⁻¹]	0.26	1.05	1.37	0.28
MAE V-E1 for y_1 [g L ⁻¹]	0.44	1.74	2.19	0.42
MAE V-E2 for y_1 [g L ⁻¹]	0.44	1.70	2.31	1.11
Output variable y_2				
MAE in calibration for y_2 [g L ⁻¹]	0.53	0.61	1.49	0.62
MAE V-E1 for y_2 [g L ⁻¹]	1.00	1.13	1.26	2.88
MAE V-I for y_2 [g L ⁻¹]	0.59	0.68	1.55	1.36
MAE V-E2 for y_2 [g L ⁻¹]	1.13	1.26	3.17	2.75

Table 5

Case study #1 – Baker's yeast growth. Average MAE for output variables y_1 and y_2 under each scenario for the different hybrid models considered in this study.

	Hybrid with mechanistic component #1	Hybrid with mechanistic component #2	Hybrid with mechanistic component #3
Output variable y_1			
MAE in calibration for y_1 [g L ⁻¹]	0.03	0.05	0.06
MAE V-I for y_1 [g L ⁻¹]	0.06	0.14	0.14
MAE V-E1 for y_1 [g L ⁻¹]	0.24	0.50	0.46
MAE V-E2 for y_1 [g L ⁻¹]	0.29	0.35	0.38
Output variable y_2			
MAE in calibration for y_2 [g L ⁻¹]	0.07	0.09	0.13
MAE V-E1 for y_2 [g L ⁻¹]	0.63	0.81	0.95
MAE V-I for y_2 [g L ⁻¹]	0.21	0.23	0.30
MAE V-E2 for y_2 [g L ⁻¹]	0.78	0.90	1.14

and initial states that fall within the range used during calibration – and two types of extrapolations, i.e., (i) extrapolation on input trajectories (V-E1) – in which the time profiles of manipulated variables differ from those used during calibration – and (ii) extrapolation on initial conditions (V-E2) – in which the initial states lie outside the range observed during calibration. These scenarios are specifically designed to evaluate the generalisation capabilities of the models beyond the calibration domain. Table 3 summarizes the estimated parameters for each mechanistic model, and Fig. 3 compares their predictive performance using a representative example (batch #1) from the validation batches.

The predictive performance of the three mechanistic models is evaluated using the MAE (Table 4). As expected, the best-performing mechanistic model (i.e., mechanistic model #1) shows consistently lower MAE values. In contrast, the other mechanistic models exhibit a

marked decline in predictive accuracy in extrapolation, particularly for different initial conditions. Overall, while a high-fidelity mechanistic model provides reliable predictions even outside the calibration domain, simplifications in the model structure compromise its performance.

As a reference for subsequent comparison, we also construct a purely data-driven RF model on the same calibration datasets, which uses the process inputs u_1 and u_2 to directly predict the output variables, y_1 and y_2 . The purely data-driven model provides good accuracy for in-domain predictions (V-I), whose performance significantly worsens in extrapolation settings (V-E1 and V-E2) (Table 4).

4.1.2. Construction of the hybrid model

We proceed with the construction of hybrid models as described in Section 2.1.2. They are built on top of each of the three mechanistic

Table 6

Case study #1 – Baker's yeast growth. Percentages of accuracy improvement of hybrid models (i.e., MAE reduction) over the corresponding standalone mechanistic models and the purely data-driven model.

	Hybrid with mechanistic model #1	Hybrid with mechanistic model #2	Hybrid with mechanistic model #3
Improvement over mechanistic models			
Output variable y_1			
MAE in calibration for y_1 [g L ⁻¹]	88 %	95 %	96 %
MAE V-I for y_1 [g L ⁻¹]	77 %	87 %	90 %
MAE V-E1 for y_1 [g L ⁻¹]	45 %	71 %	79 %
MAE V-E2 for y_1 [g L ⁻¹]	34 %	79 %	83 %
Output variable y_2			
MAE in calibration for y_2 [g L ⁻¹]	87 %	85 %	91 %
MAE V-E1 for y_2 [g L ⁻¹]	37 %	28 %	25 %
MAE V-I for y_2 [g L ⁻¹]	64 %	66 %	81 %
MAE V-E2 for y_2 [g L ⁻¹]	31 %	29 %	64 %
Improvement over data-driven model			
Output variable y_1			
MAE in calibration for y_1 [g L ⁻¹]	77 %	62 %	54 %
MAE V-I for y_1 [g L ⁻¹]	79 %	50 %	50 %
MAE V-E1 for y_1 [g L ⁻¹]	43 %	-19 %	-9 %
MAE V-E2 for y_1 [g L ⁻¹]	74 %	68 %	66 %
Output variable y_2			
MAE in calibration for y_2 [g L ⁻¹]	89 %	85 %	79 %
MAE V-E1 for y_2 [g L ⁻¹]	78 %	72 %	67 %
MAE V-I for y_2 [g L ⁻¹]	85 %	83 %	78 %
MAE V-E2 for y_2 [g L ⁻¹]	72 %	67 %	59 %

components. For each hybrid model, RFs are obtained based on the same calibration datasets adopted in Section 4.1.1. The data-driven inputs are the same process variables used in the mechanistic component (i.e., u_1 and u_2), together with the outputs predicted by the mechanistic component (i.e., y_1 and y_2) over time. RFs hyperparameters are optimised via five-fold cross-validation, with a selected number of 100 trees for all cases. The same validation settings as in Section 4.1.1 are leveraged to assess the benefits of the hybrid models (i.e., V-I, V-E1, and V-E2).

Quantitative analysis of the MAE values reported in Table 5 and percentages of improvement over the standalone mechanistic and data-driven models in Table 6 clearly highlights the advantages of hybridisation, especially over mechanistic models #2 and #3. For instance, in the most challenging extrapolation scenario (V-E2), the hybrid model built on mechanistic component #3 reduces the MAE for y_2 from 3.17 g L^{-1} (purely mechanistic) and 2.75 g L^{-1} (purely data-driven) down to 1.14 g L^{-1} , corresponding to a 64 % and 59 % reduction, respectively. Similarly, in the in-domain setting (V-I), the hybrid model built on

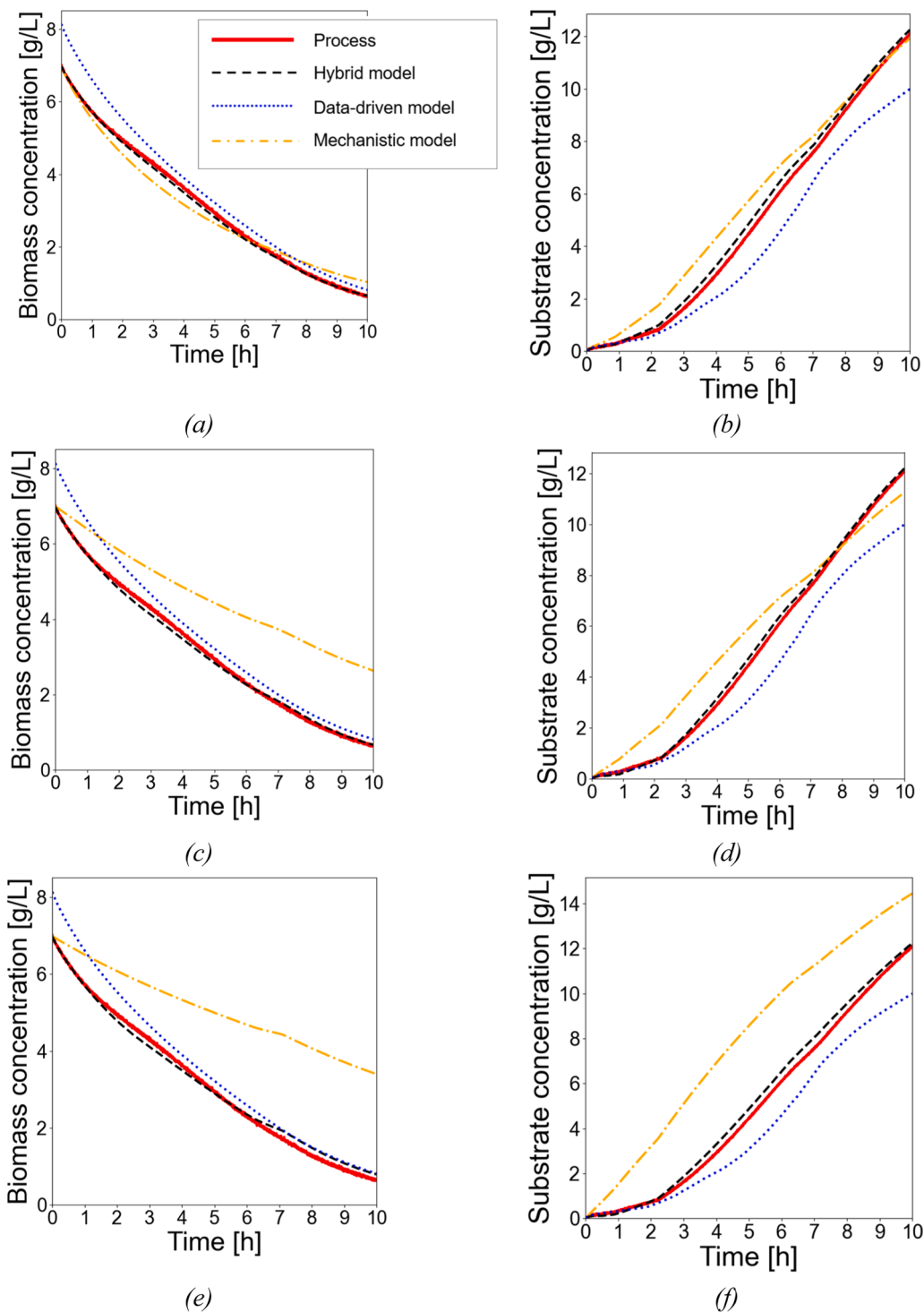


Fig. 4. Case study #1 – Baker's yeast growth. Comparison of model predictions for biomass concentration y_1 and substrate concentration y_2 on validation batch #1 (a–b) hybrid model based on mechanistic component #1, (c–d) on mechanistic component #2, (e–f) on mechanistic component #3. In each panel, predictions from the mechanistic model, the purely data-driven model, and the hybrid model are compared against the reference process data.

mechanistic component #2 lowers the MAE for y_1 by 87 % compared to the corresponding mechanistic model (from 1.05 to 0.14 g L⁻¹), and by 50 % compared to the purely data-driven one (from 0.28 to 0.14 g L⁻¹). These results confirm that even limited mechanistic knowledge, when suitably combined with flexible data-driven components, can lead to models with robust predictive capabilities across a broad range of scenarios.

Fig. 4 provides a visual comparison of the hybrid model predictions under all three configurations, benchmarked against the respective mechanistic and the purely data-driven models, and using the same validation batch as in Section 4.1.1.

4.1.3. Effect of the amount of information available

In this section, we analyse how the amount of calibration data impacts the predictive capability of the hybrid model. To this end, we repeat the procedure described in Sections 4.1.1 and 4.1.2, using mechanistic model #1 and mechanistic model #3 as mechanistic components, which represent the two extremes in terms of predictive capabilities (i.e., the best and the worst performances, respectively). The analysis is conducted by reducing the available calibration dataset to 10 % (Table 7) of its original size, and evaluating the resulting accuracy under the same validation scenarios.

Fig. 5 illustrates the predictive performance for the two hybrid models and the data-driven model, using the same validation batch as in Sections 4.1.1 and 4.1.2, when calibration is performed with a decreasing-size dataset that was obtained through random selection. The mechanistic models are not included in the comparison in Fig. 5, given that their performance does not significantly vary with the amount of calibration data.

Results show that a substantial reduction in data availability to only 10 % of the original dataset size does not lead to a notable decline in predictive performance. Compared to the purely data-driven model, both hybrid models consistently demonstrate superior robustness to reductions in calibration data, especially the one built upon mechanistic component #1, which embeds the higher level of mechanistic knowledge. The superior resilience of hybrid model built upon mechanistic component #1 proves the pivotal role played by the underlying mechanistic component in determining the overall performance of hybrid frameworks, particularly when experimental data are limited.

Key findings from the analysis of hybrid model performance under different levels of mechanistic knowledge and information available can be summarized as follows.

- The accuracy of the mechanistic component strongly influences the overall hybrid model performance. While hybrid models consistently outperform purely data-driven models across all cases, the magnitude of improvement is smaller when the mechanistic component poorly represents the system (e.g., mechanistic model #3);

- The effect of the dataset reduction (down to 10 % of the original dataset size) is more evident for the hybrid model that incorporates the poorly performing mechanistic component (i.e., mechanistic model #3), where the limited information contained in a small dataset cannot fully compensate for the inadequacy of the mechanistic model;
- The advantages of hybrid models can be immediately appreciated under in-domain conditions (V-I). In this setting, all models, including purely mechanistic and data-driven models, show relatively good predictive accuracy. However, hybrid models still retain a consistent edge. For example, when leveraging the full calibration dataset (100 batches), the hybrid configuration based on mechanistic component #1 achieves an MAE of 0.06 g L⁻¹ for y_1 compared to 0.26 g L⁻¹ for the mechanistic model and 0.28 g L⁻¹ for the purely data-driven one; similarly, the MAE for y_2 is 0.21 g L⁻¹, versus 0.59 (mechanistic) and 1.36 (data-driven).
- The benefits of hybrid models become more evident when the prediction task involves extrapolation, i.e., moving beyond the conditions considered for calibration. In the most challenging validation scenario (V-E2) and still considering case of full calibration dataset, the hybrid configuration built upon mechanistic component #1 (which embeds the most informative process description) shows very low MAE values, corresponding to improvements of 34 % and 74 % for y_1 , and 31 % and 72 % for y_2 , when compared to the purely mechanistic and data-driven models, respectively. Remarkably, even the hybrid model built upon mechanistic component #3 – despite relying on more limited process knowledge – delivers substantial gains. For y_1 , the MAE is reduced by 64 % relative to the mechanistic model and 66 % relative to the data-driven one. For y_2 , the corresponding reductions are 64 % and 59 %.

4.2. Ion-exchange chromatography

4.2.1. Mechanistic model calibration and validation

Focusing on the main component, i.e., the mAb, mechanistic model parameters of both models are estimated using four of the five available experiments (Section S.2 in the Supplementary Material) – which represents a very low amount of data for model calibration. This procedure is repeated by alternatively excluding one experiment, ensuring that each one is used for validation once. The mean parameter values reported in Table 8 are computed as the arithmetic average over all estimation runs.

The accuracy of the two mechanistic models is quantitatively evaluated using the MAE and the normalized peak error (PE), which is defined as:

$$PE = \frac{|\hat{t}_{peak} - t_{exp, peak}|}{\Delta t} \quad (25)$$

Table 7

Case study #1 – Baker's yeast growth. Average performance metrics for output variables y_1 and y_2 under each scenario for the different models (i.e., mechanistic, hybrid and data-driven) considered in this study when the available dataset is reduced to 10 %.

	Mechanistic model #1	Mechanistic model #3	Hybrid with mechanistic component #1	Hybrid with mechanistic component #3	Data-driven model
MAE in calibration for y_1 [g L ⁻¹]	0.31	1.76	0.07	0.17	0.35
MAE in calibration for y_2 [g L ⁻¹]	0.68	1.65	0.16	0.26	0.90
MAE V-I for y_1 [g L ⁻¹]	0.28	1.65	0.16	0.32	1.06
MAE V-I for y_2 [g L ⁻¹]	0.62	1.26	0.37	0.49	3.31
MAE V-E1 for y_1 [g L ⁻¹]	0.43	2.57	0.33	0.77	1.08
MAE V-E1 for y_2 [g L ⁻¹]	1.12	1.30	0.91	1.11	4.13
MAE V-E2 for y_1 [g L ⁻¹]	0.49	2.84	0.35	0.71	1.46
MAE V-E2 for y_2 [g L ⁻¹]	1.15	2.67	0.90	1.34	2.76

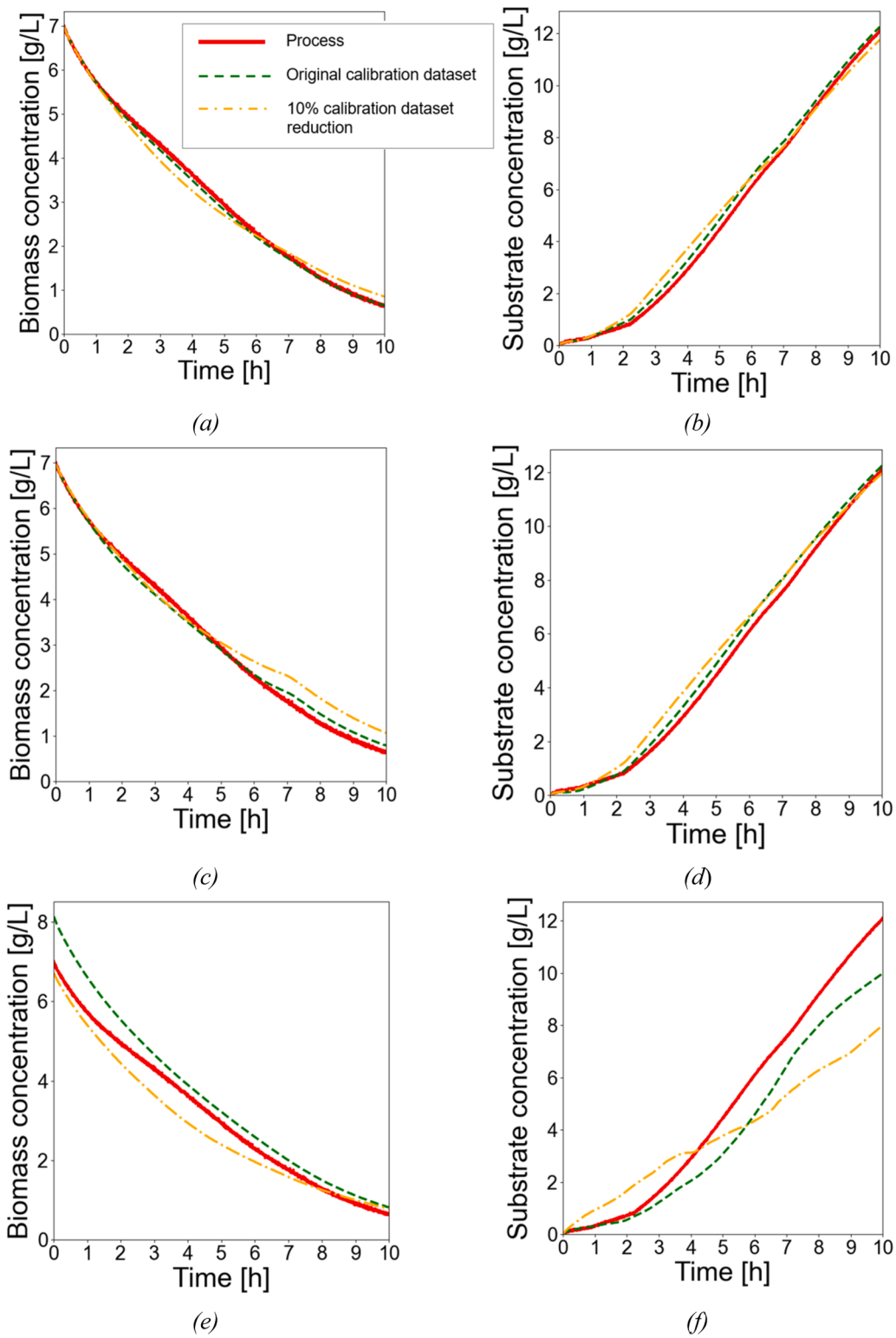


Fig. 5. Case study #1 – Baker's yeast growth. Comparison of model predictions under reduced calibration data conditions (10 % of the original dataset). Each panel shows the predicted trajectories against the ground truth for output variables y_1 and y_2 , using different model configurations: (a–b) hybrid model based on mechanistic component #1; (c–d) hybrid model based on mechanistic component #3; (e–f) purely data-driven model.

Table 8

Case study #2 – Ion-exchange chromatography. Estimated values of model parameters for the two mechanistic models of different levels of complexity.

Parameter	Units	Value
Langmuir adsorption isotherm		
$k_{a, mAb}$	s^{-1}	1.00×10^{-2}
$k_{d, mAb}$	s^{-1}	0.896
$q_{max, mAb}$	–	2.16
ASMA isotherm		
$k_{a, mAb}$	$(m^3 mol^{-1})^{1/2} s^{-1}$	9.29×10^{-6}
$k_{d, mAb}$	$(m^3 mol^{-1})^{1/2} s^{-1}$	2.53×10^{-8}
u_{mAb}	–	4.10
σ_{mAb}	–	10.94
K_P, mAb	$m^3 mol^{-1}$	-5.04×10^{-3}
K_1, mAb	$m^3 mol^{-1}$	-3.62×10^{-4}

where $\hat{t}_{peak}$ and $t_{exp, peak}$ are the time at which the maximum value of predicted output and the experimental counterpart (respectively) appear; Δt is the time span in which the experimental output concentration is non-zero. This metric serves as an effective indicator of the accuracy in forecasting the time of peak concentration, which is a critical information for the specific application (Guiochon et al., 2006).

Fig. 6 illustrates the predictive performance of the calibrated mechanistic models (parameters as in Table 8) across all available datasets, while accuracy metrics are reported in Table 9. For both models, predictions are poor, suggesting that the available model structures cannot describe well the underlying physical phenomena. Notably, the LKM coupled with the simple Langmuir adsorption isotherm exhibits substantial limitations under conditions with varying salt concentrations in the mobile phase, which induce complex binding interactions, as seen in datasets A, C, and E. In these cases, the model fails to capture delayed elution peaks, as indicated by the high *PE* values. This is consistent with the fact that, conversely, datasets B and D are characterized by constant salt concentration and lack of peak distortion, which reduce nonlinear binding effects and render the system dynamics more consistent with the assumptions of the simple Langmuir adsorption isotherm.

In contrast, the ASMA isotherm demonstrates enhanced descriptive capabilities, particularly in capturing the temporal characteristics of elution peaks, as reflected by the lower *MAE* and *PE* values in datasets A, C, and E. Nevertheless, the performance of the ASMA-based model remains dataset-dependent, indicating that improvements are needed to ensure robust predictive accuracy across a broader range of operating conditions.

A purely data-driven RF model is also constructed. The model uses the process inputs (i.e., conductivity κ , concentration of modulator c_1 , and initial conditions) to directly predict the outlet mAb concentration, without relying on any mechanistic prior. Calibration results and validation performance of the purely data-driven model are reported in the Supplementary Material (Section S.2). The purely data-driven model, while achieving moderate accuracy for datasets B and D, underperforms in complex conditions like those in datasets A, C, and E, where the lack of mechanistic priors prevents it from generalizing well to nonlinear retention behaviour (Table 9).

4.2.2. Construction of the hybrid model

We then proceed with the construction of hybrid models following the methodology described in Section 2.1. Two hybrid configurations are developed, each built upon a different mechanistic component, i.e., one based on the LKM coupled with the Langmuir adsorption isotherm, and the other on the LKM coupled with the ASMA isotherm. For both

configurations, RF regressors are constructed on four of the five available experimental datasets, with a leave-one-out cross-validation scheme to ensure robust model assessment. The inputs to the RF consist of the outlet concentration and the internal states predicted by the mechanistic component (i.e., c_{mAb} and q_{mAb}), together with the conductivity profile κ . RF hyperparameters are optimised via five-fold cross-validation, and a forest size of 100 trees is selected in both cases.

For sake of conciseness, only cross-validation results are shown in Figs. 7 and 8, which illustrate the predictive performance of the hybrid models across the five experimental datasets built on the LKM coupled with Langmuir adsorption isotherm and the ASMA isotherm, respectively. It is important to state that both hybrid and purely data-driven (Section 4.2.1) configurations exhibit excellent performance in calibration, consistently achieving low error values across all datasets.

Metrics used for the assessment of the predictive capabilities of the constructed models both in calibration and validation (i.e., *MAE* and *PE* values) are reported in Table 10. The hybrid model built on the Langmuir isotherm performs reasonably well only for datasets B and D, where system behaviour appears simpler and closer to the assumptions of the Langmuir model. Instead, it fails to correct the poor predictions of the mechanistic model for datasets A, C, and E, where peak broadening and delayed elution dominate the chromatographic response, as confirmed by high *PE* values.

In contrast, the hybrid model based on the ASMA isotherm achieves significantly better accuracy across most scenarios, with notable improvements in datasets A, C, and D. For example, in dataset A, the *MAE* drops from 1.22 g L^{-1} (mechanistic) and 1.19 g L^{-1} (purely data-driven) to 0.28 g L^{-1} , which corresponds to a reduction of approximately 77 % and 76 %, respectively. Large improvement is also evident in dataset C, where the hybrid-ASMA model reaches a *MAE* of 0.57 g L^{-1} , reducing the *MAE* by 69 % and 41 % compared to the purely data-driven and mechanistic models, respectively.

Although it still exhibits limitations in dataset E (cr. Tables 9 and 10), the model succeeds in capturing the shape and timing of the elution peak more effectively, indicating that the more flexible ASMA structure allows for better integration with the data-driven correction.

Key outcomes from this analysis can be summarised as follows:

- The hybrid model based on the ASMA isotherm shows the best overall predictive performance across the majority of validation cases, with up to 77 % reduction in *MAE* compared to both the purely data-driven and the mechanistic models (dataset A). The only exception is dataset E, where the *MAE* is higher when using the hybrid model. It should be noted, however, that the peak time and shape prediction is improved. In scenarios characterized by more complex elution behaviours (e.g., datasets A and C), peak predictions are improved in both magnitude and timing;
- The hybrid model based on the Langmuir adsorption isotherm performs well only in simple scenarios (i.e., dataset B and D), while fails in presence of delayed elution peaks. The trend is also reflected by the purely data-driven model, that shows competitive performance only when datasets B and D are used for validation;
- With specific concern on dataset E, the purely mechanistic ASMA model outperforms both the hybrid and data-driven alternatives, indicating that hybridisation may not always be beneficial, e.g., when the mechanistic component already captures the essential dynamics and in the presence of limited calibration datasets.

To provide an explanation to the reduced performance observed on the hybrid model predictive capability when datasets A, C and E are used in validation, a PCA-based analysis is proposed and described in Appendix A.1.

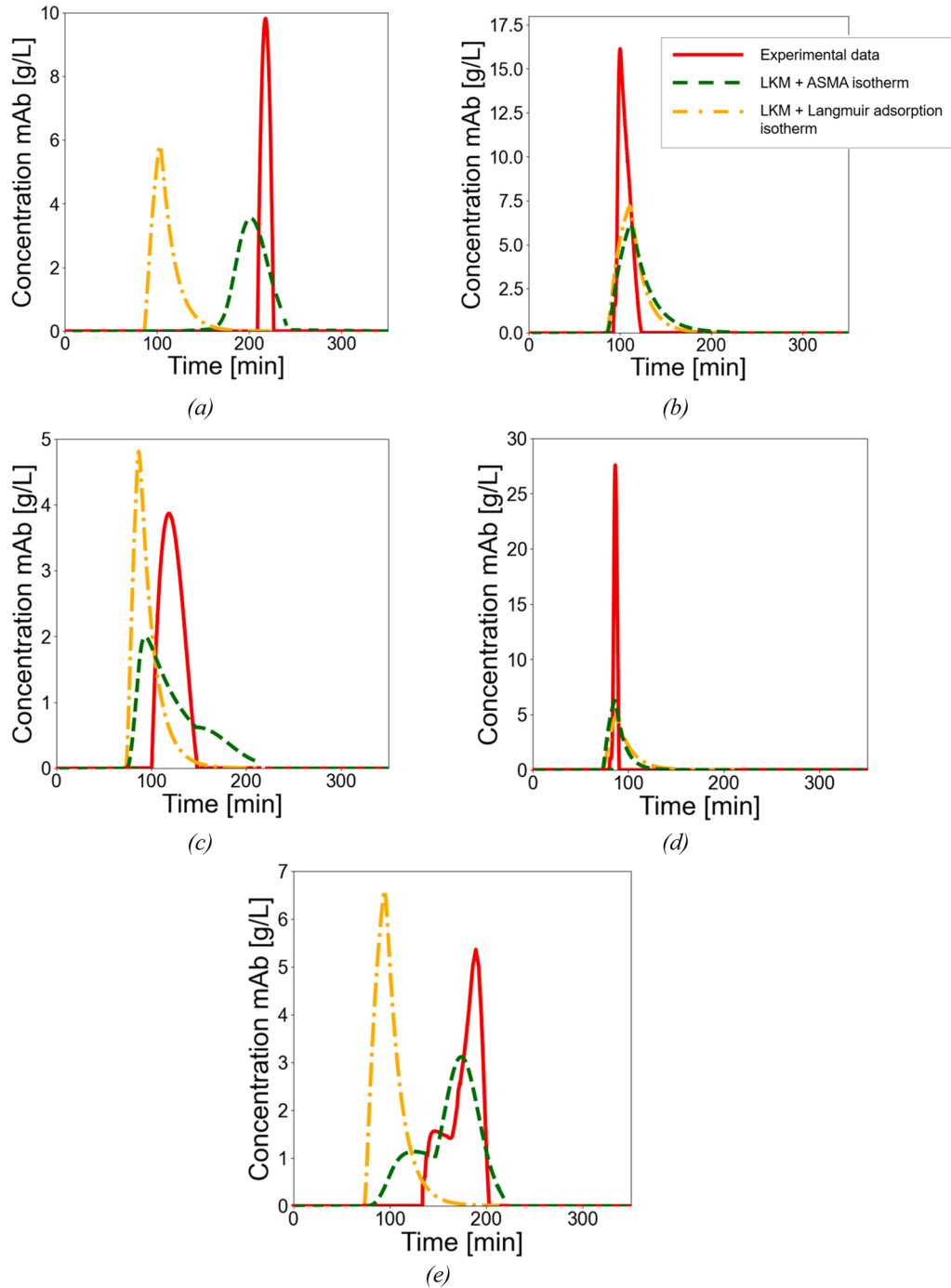


Fig. 6. Case study #2 – Ion-exchange chromatography. Performance of the two mechanistic models with respect to output concentration of mAb for the available experimental runs: (a) A, (b) B, (c) C, (d) D, and (e) E.

Table 9

Case study #2 – Ion-exchange chromatography. MAE and PE under each scenario for the different mechanistic models and purely data-driven model considered in this study.

	<i>LKM + Langmuir adsorption isotherm</i>	<i>LKM + ASMA isotherm</i>	<i>Data-driven model</i>
Average MAE in calibration [g L^{-1}]	1.33	1.06	1.11
MAE – Dataset A [g L^{-1}]	1.87	1.22	1.19
MAE – Dataset B [g L^{-1}]	1.24	1.51	1.82
MAE – Dataset C [g L^{-1}]	1.24	0.91	1.83
MAE – Dataset D [g L^{-1}]	1.14	1.04	0.81
MAE – Dataset E [g L^{-1}]	2.32	0.68	0.86
Average PE in calibration [g L^{-1}]	1.77	0.33	0.02
PE – Dataset A [-]	6.52	0.81	0.03
PE – Dataset B [-]	0.35	0.38	0.05
PE – Dataset C [-]	0.67	0.52	1.62
PE – Dataset D [-]	0.04	0.06	0.05
PE – Dataset E [-]	1.25	0.04	0.32

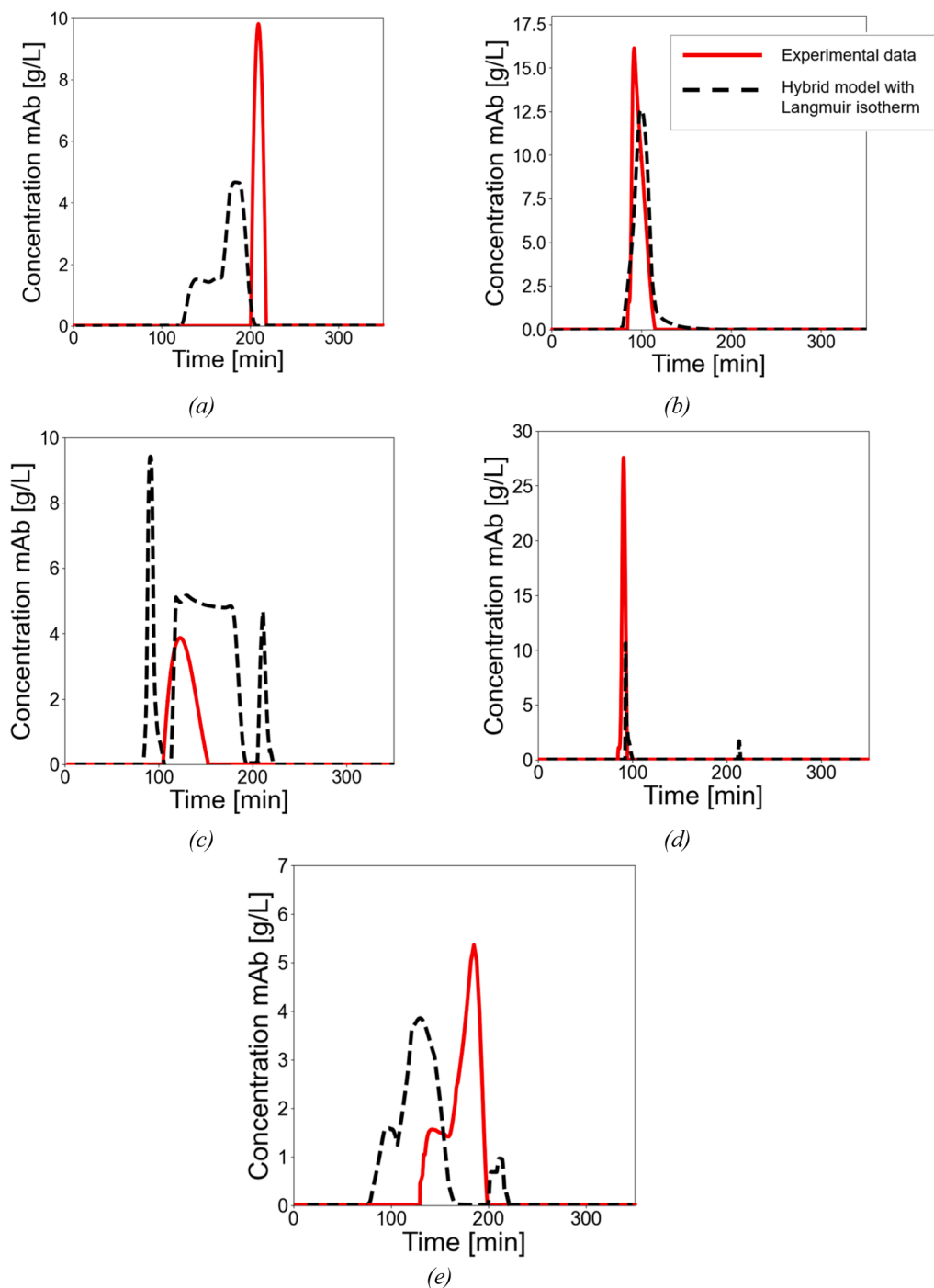


Fig. 7. Case study #2 – ion-exchange chromatography. Cross-validation results. Performances of the hybrid model built on LKM coupled with Langmuir adsorption isotherm (black dashed lines) with respect to output concentration of mAb for the available experimental runs (red line): (a) A, (b) B, (c) C, (d) D, and (e) E.

4.3. Discussion

The results confirm the potential of hybrid modelling strategies to enhance the predictive performance of standalone mechanistic and data-driven models, based on the level of mechanistic knowledge and the amount of information available for model training. However, the benefit of hybridisation is not unconditional and depends on several interacting factors. Regarding the impact of the two factors analyzed on hybrid model performance, namely (i) mechanistic model accuracy and (ii) amount of information available for model calibration, the following observations can be drawn:

(i) Hybrid modelling is not always advantageous. The benefits depend on the specific characteristics of the system under study and the available information. The accuracy of the mechanistic component plays a pivotal role in shaping the hybrid performance: when it provides a good approximation of the process (as observed for mechanistic model #1, the ASMA-based chromatography model, and the urethane case study) hybrid structure significantly improves prediction accuracy and generalisation, even under extrapolative conditions. However, when the mechanistic component is too simplistic or misleading (as for mechanistic model #3 or Langmuir-based chromatography model under salt-varying conditions), and

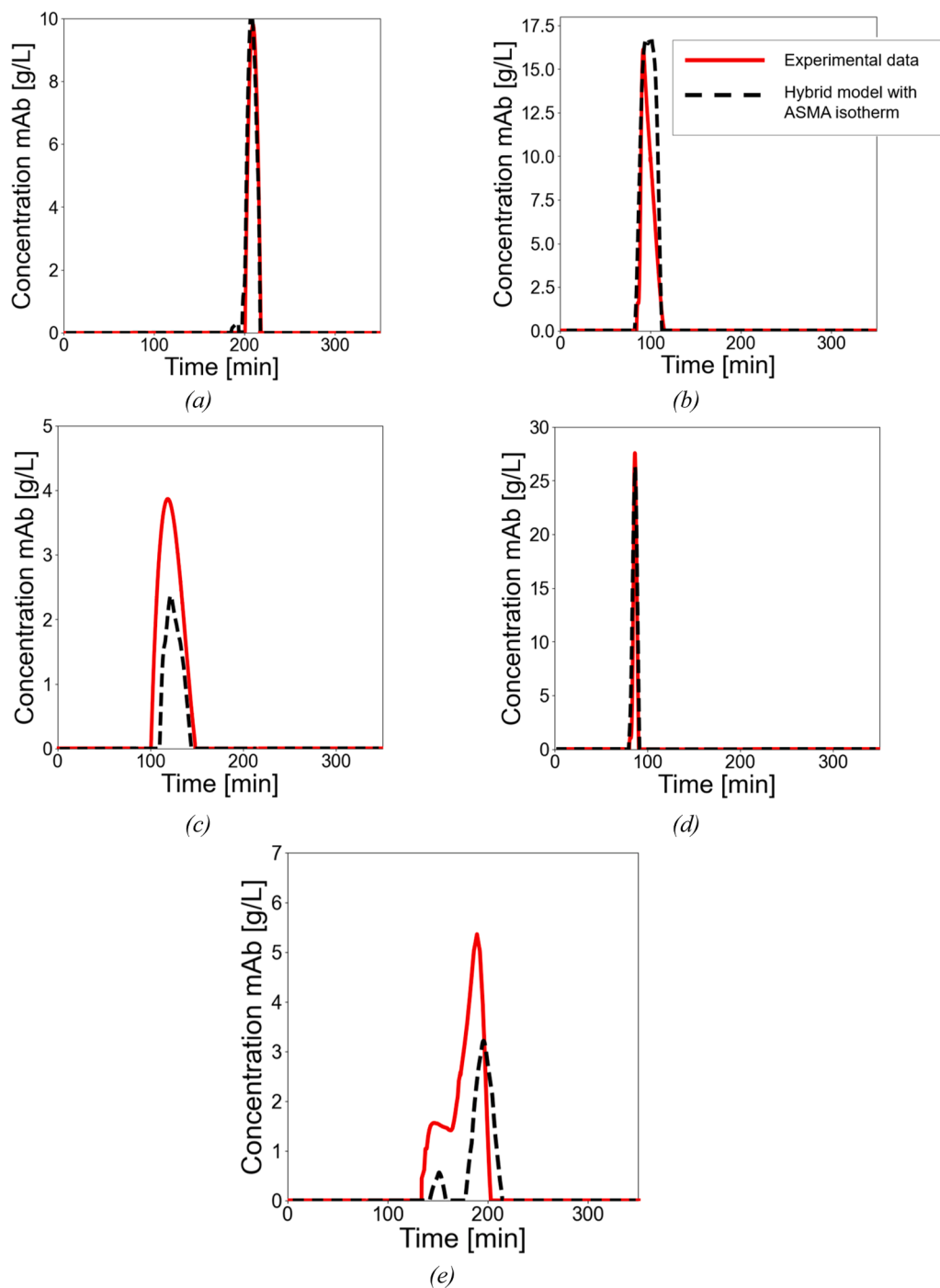


Fig. 8. Case study #2 – Ion-exchange chromatography. Cross-validation results. Performances of the hybrid model built on LKM coupled with ASMA isotherm (black dashed lines) with respect to output concentration of mAb for the available experimental runs (red line): (a) A, (b) B, (c) C, (d) D, and (e) E.

the data are insufficient to compensate, the added complexity of hybridisation cannot be justified.

(ii) The amount of information available for model training critically influences the predictive capability of the hybrid model. Our analysis shows that hybrid models maintain good predictive performance even in low-data regimes, provided that the mechanistic component is sufficiently informative. In contrast, purely data-driven models are highly sensitive to the amount of data available for the calibration and may fail to extrapolate. However, when both data and mechanistic insights are scarce, hybrid models struggle as well. Even, results from the chromatography case (dataset E) show that in some scenarios, the standalone mechanistic model can outperform the

hybrid one. Finally, the urethane case study (Supplementary material) shows consistent hybrid model performance, even when the noise in the data increases.

The main conclusions drawn from the discussion are resumed in [Table 11](#).

5. Conclusions

In this work, we proposed a methodology to develop hybrid model configurations, focusing on the interplay between mechanistic model fidelity and the amount of information available for model training. A

Table 10

Case study #2 – Ion-exchange chromatography. MAE for the different hybrid models considered in this study. Percentages of improvement of hybrid model over corresponding standalone mechanistic models are also reported.

	Hybrid with Langmuir isotherm	Improvement over mechanistic (Langmuir)	Hybrid with ASMA isotherm	Improvement over mechanistic (ASMA)
Average MAE in calibration [g L ⁻¹]	0.43	68 %	0.37	65 %
MAE – validation Dataset A [g L ⁻¹]	2.09	–12 %	0.28	77 %
MAE – validation Dataset B [g L ⁻¹]	0.93	25 %	1.10	27 %
MAE – validation Dataset C [g L ⁻¹]	2.62	–111 %	0.57	37 %
MAE – validation Dataset D [g L ⁻¹]	0.94	18 %	0.37	64 %
MAE – validation Dataset E [g L ⁻¹]	1.97	15 %	0.89	–31 %
Average PE in calibration [-]	0.01	99 %	0.01	97 %
PE – validation Dataset A [-]	1.52	77 %	0.15	81 %
PE – validation Dataset B [-]	0.26	26 %	0.26	32 %
PE – validation Dataset C [-]	0.66	0 %	0.09	82 %
PE – validation Dataset D [-]	0.22	–450 %	0.03	55 %
PE – validation Dataset E [-]	0.79	37 %	0.09	–134 %

Table 11

Summary of the key findings of this study.

Mechanistic model accuracy	Data information content		Hybrid model advantage	Key observations
High	High	Moderate		Hybrid model slightly improves predictions, mostly in extrapolation.
High	Low	Strong		Mechanistic knowledge compensates for limited data; hybridization beneficial.
Low	High	Moderate		Data compensate partially for structural mismatch.
Low	Low	Weak/None		Hybridization not justified: limited improvement.

central objective was to assess the conditions under which a hybrid approach offers real advantages over purely mechanistic or data-driven alternatives. Results suggest that the effectiveness of hybrid modelling critically depends on both the quality of the mechanistic component and the availability of data. When mechanistic knowledge offers a reliable process representation, hybrid models can outperform purely data-driven alternatives, particularly in data-scarce settings. However, when the mechanistic component is overly simplistic or misleading, and data are insufficient to compensate, the added complexity of hybridisation may not yield benefits.

Future work could explore the application of the methodology to other case studies, including processes from different industries, in order to assess its generalizability and to evaluate how hybrid modelling strategies perform under diverse process and data availability conditions.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the generative artificial intelligence program *Lucrez-IA* by University of Padova was used to produce a first draft of the abstract. The authors reviewed and edited the abstract content as needed and take full responsibility for the content of the publication.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.compchemeng.2025.109536](https://doi.org/10.1016/j.compchemeng.2025.109536).

Appendix A

To understand why certain datasets yield better predictive performance when used for validation (in both the hybrid modelling framework and purely-data driven alternative), we analyse their similarities through principal component analysis (PCA) (Montgomery, 2013). In general, PCA serves to summarize the information of the dataset matrix $\mathbf{D}$ [$N_{sp} \times M$] of the N_{sp} observations of M measured variables by projecting them onto a new coordinate system of B principal components (PCs), aiding visualization and graphical interpretation. To avoid the scaling effect of different orders of magnitude of the M variables, $\mathbf{D}$ is autoscaled, i.e., data are mean-centered and scaled to unit variance (Wise and Gallagher, 2006). This representation of $\mathbf{D}$ in PCA is given by:

$$\mathbf{D} = \mathbf{S}\mathbf{R}^T + \mathbf{E}_D, \quad (\text{A.1})$$

CRediT authorship contribution statement

Margherita Geremia: Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Tiziana Marella:** Writing – review & editing, Software, Methodology, Formal analysis, Conceptualization. **Elaheh Ardalani:** Writing – review & editing, Resources, Conceptualization. **Samira Beyramysoltan:** Writing – review & editing, Resources, Conceptualization. **Sayantan Chattoraj:** Writing – review & editing, Resources. **Pierantonio Facco:** Writing – review & editing, Methodology, Conceptualization. **Massimiliano Barolo:** Writing – review & editing, Methodology, Conceptualization. **Fabrizio Bezzo:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: E. A., S.B., and S.C. are employees of GSK.

where $\mathbf{S}$ [$N_{sp} \times M$] is the score matrix, $\mathbf{R}$ [$M \times B$] is the matrix of loadings, and $\mathbf{E}_D$ [$N_{sp} \times M$] is the matrix of residuals. The number of PCs, B , is selected according to the “eigenvalue-greater-than-one” rule (Kaiser, 1991).

The PCA-based analysis proceeds as follows:

- (i) for each experiment (i.e., each batch), arrange the corresponding data in the variable-wise unfolded matrix $\mathbf{D}$ (Wold et al., 1987) - the full dataset matrix $\mathbf{D}$ thus contains all N_{sp} observations across all experiments. Apply PCA to project the observations of each experiment onto the score space, using the loadings on the principal components as coordinates, and compute the centroid for each experiment to represent the entire batch as a single point in the score space (Fig. A.1a);
- (ii) compute the Euclidian distances between the centroids of all datasets to determine their similarity (Fig. A.1b);
- (iii) identify the closest pair of centroids, i.e., the pair characterized by the minimum Euclidian distance (Fig. A.1c);
- (iv) among the pair identified in (iii), the dataset that is closer to all other datasets, i.e., its mean Euclidian distance with respect to the other centroids is the lowest, indicates it is less unique (Fig. A.1d). When used for validation, this dataset ensures that no relevant information is lost.

Note that the centroid-based representation assumes that the time profiles within each experiment are not dispersed. In cases where time dynamics are highly dispersed, the centroid may be less representative of the full trajectory in the score space.

The main scope of this analysis is to justify the different performances of the hybrid and the purely data-driven models when the validation dataset is changed. Furthermore, it provides a fast assessment of the dissimilarity between datasets, and their capability to map a representative design space. Note that in real life, it is likely that all available datasets would be used for model calibration: when data are scarce there is no reason to remove any of them (unless outliers are present).

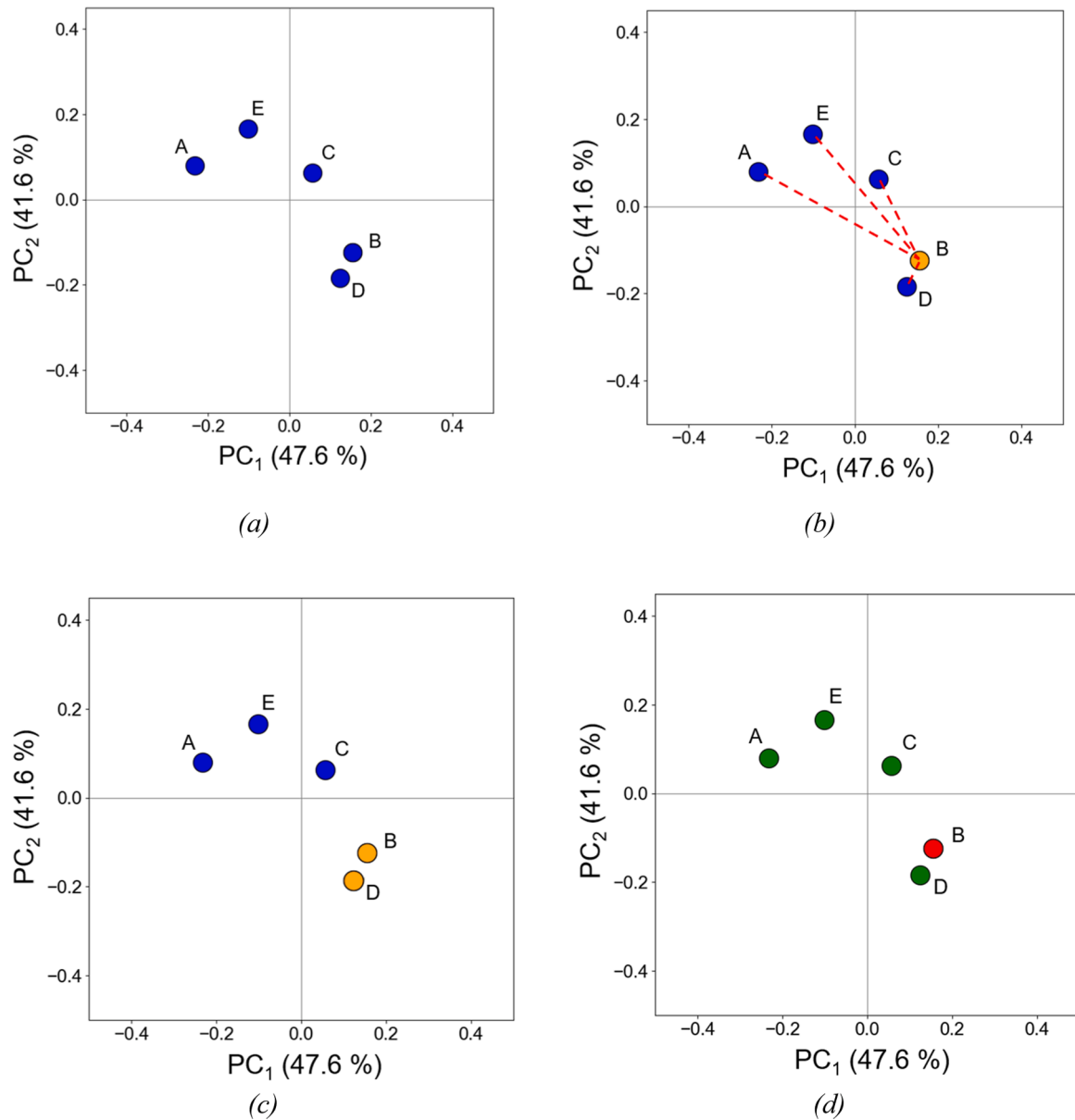


Fig. A.1. PCA for dataset selection in Case study #2. (a) Projection of experimental datasets in score space. (b) Euclidean distances between dataset B centroid and the others. (c) Closest centroids identified (B and D). (d) Calibration (A, C, D, E) and validation (B) datasets. Explained variance for each PC is shown in brackets.

For the presented case study, PCA results demonstrate that dataset B is very similar to dataset D (Fig. A.1c). This is coherent with the fact that the best validation performances of the hybrid model based on the LKM coupled with the Langmuir adsorption isotherm and the purely data-driven model are achieved using datasets A, C, D, E for calibration, and dataset B for validation.

Data availability

The data that has been used is confidential.

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